

EQUIFLOW QA ASSESSMENT

AFTER REPORT: 100% Test Case Pass Rate & Production Deployment

STATUS: ' 19/19 PASSED (MongoDB Atlas Cloud & Render Live Host)

Generated: 2026-08-15 | Live Host: <https://tactive.onrender.com/>

1. Executive Resolution & Architecture Summary

All injected regressions and test assertion mismatches have been resolved. The database was successfully migrated from SQLite to MongoDB Atlas Cloud Cluster. The entire stack has been deployed live to Render at <https://tactive.onrender.com/> with 100% passing automated test coverage.

2. AFTER Test Execution Matrix (100% PASS)

Category	Test Scenario & Expectation	After Outcome
1. Happy Path	GET /api/assets -> Seeded inventory list returned	PASS [HTTP 200] '
1. Happy Path	GET /api/health -> Health status check	PASS [HTTP 200] '
1. Happy Path	VIP User auto-confirmed reservation (No Admin queue)	PASS [HTTP 201] '
1. Happy Path	Standard User reservation -> Enters PENDING queue	PASS [HTTP 201] '
1. Happy Path	Admin can approve PENDING reservation	PASS [HTTP 200] '
1. Happy Path	Admin can reject PENDING reservation	PASS [HTTP 200] '
1. Happy Path	User can cancel own PENDING reservation	PASS [HTTP 200] '
2. Business Rules	CONFLICT: Overlapping dates rejected (conflictEngine)	PASS [HTTP 409] '
2. Business Rules	QUOTA: Standard user blocked at >3 active bookings	PASS [HTTP 400] '
2. Business Rules	BLACKOUT: Booking blocked during admin maintenance	PASS [HTTP 409] '
2. Business Rules	PENALTY (VIP): 3 days overdue -> \$45 (50% VIP discount)	PASS [HTTP 200] '
2. Business Rules	PENALTY (Standard): On-time return = \$0 penalty	PASS [HTTP 200] '
3. Validation	REJECT: Inverted dates (start > end)	PASS [HTTP 409] '
3. Validation	REJECT: Booking duration > 14 days limit	PASS [HTTP 409] '
3. Validation	REJECT: Non-existent asset ID returns 404	PASS [HTTP 404] '
3. Validation	REJECT: Unauthenticated request returns 401	PASS [HTTP 401] '
3. Validation	REJECT: Standard user cannot create blackouts	PASS [HTTP 403] '
3. Validation	REJECT: Missing required fields (Zod 400)	PASS [HTTP 400] '
3. Validation	REJECT: Invalid date format (Zod 400)	PASS [HTTP 400] '

3. Captured Terminal Output (100% Green Run)

